Task-directed post-hoc protection

Task directions and finite construction. Let W_l be a frozen expert matrix and $W_{0,l}$ its public-base counterpart. We take the first 32 right singular vectors V_l of $W_l - W_{0,l}$ for 74 logical matrices: Q/K/V/O/A/B in 12 blocks, the flattened patch map, and the stored visual projection. For a shared residual-coordinate permutation P, let $E_l(P)$ be the actual finite matrix change and define

$$T(P) = \sum_l \frac{\|E_l(P)\|_F^2}{\|W_l\|_F^2}, \qquad I(P) = \sum_l \frac{\|E_l(P)V_l\|_F^2}{\|W_l\|_F^2}, \qquad O(P) = T(P) - I(P).$$

Starting from identity, we maximize $F(P) = O(P) - I(P)$ by disjoint coordinate swaps. At each step we select the largest exact positive finite gain above 10^{-10}, with lexicographic ties, and stop after at most 384 swaps. This is a fixed greedy heuristic, without a guarantee of global optimality or an absolute bound on I. Construction uses only expert/base parameters; no images, partner, classification scores, or expert training are used.

Coordinated function preservation. Q/K/V/A read residual coordinates and use $W_l P^\top$; O/B write residual coordinates and use PW_l. Patch outputs, embeddings, all LayerNorm gains/biases, and the visual readout move with the same coordinates. Internal attention-head and MLP-hidden coordinates stay fixed. This known symmetry preserves the real-arithmetic function; complete native FP32 predictions and relative class gaps are checked separately. The task directions remain in the original parameter coordinates and are not assumed to be a functional nullspace.

Merging and recovery scope. The original AGA changes attention/MLP parameters and leaves the patch map unchanged. For our involution $P^2 = I$, a self-average patch map is $(I + P)W_t/2$. With 384 disjoint swaps this projects patch outputs to 384 coordinates. For an actual partner, $K = (PW_t + W_p)/2$ and $N = (I - P)/2$ satisfy

$$NK = \tfrac{1}{2}N(W_p - W_t) = \tfrac{1}{2}N(T_p - T_t).$$

The actual merged map can remain full rank; this relation is not a classification-failure guarantee. We therefore evaluate real TA 0.5 and unchanged original AGA, with target-only and both-expert scopes and both task accuracies. Known-family public patch matching is a separate recovery permission setting. The distance-only order is a fixed ablation; standard symmetry, SVD and greedy tools are not individually claimed as new contributions.